Year 12 Maths - Coordinate geometry

Similar questions - Tangents to circles

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A circle has equation $x^2 + y^2 - 4x + 2y - 5 = 0$. The point $P(3,2)$ lies on the circle.

(a) Find the centre and the exact radius of the circle. **[3]**

(b) Find the equation of the tangent to the circle at P, giving your answer in the form $x + by = c$. **[3]**

(c) Find the area of the triangle bounded by this tangent and the two coordinate axes. **[2]**

(d) A second tangent to the circle is parallel to the tangent at P. By considering the point of the circle that is opposite P, find the equation of this second tangent. **[3]**

2 The line $y = 2x + k$, where k is a positive constant, is a tangent to the circle $x^2 + y^2 = 20$.

(a) Show that $x^2 + \frac{4k}{5}x + \frac{k^2-20}{5} = 0$ at any point where the line meets the circle. **[2]**

(b) Hence find the value of k. **[3]**

(c) Find the coordinates of the point where the line touches the circle. **[2]**

(d) Show that the line through the origin and the point of contact is perpendicular to the tangent, and explain why this must happen. **[2]**

Question 1

(a) $(x-2)^2+(y+1)^2=10$. Centre $(2,-1)$, radius $\sqrt{10}$

(b) Radius gradient $\dfrac{2-(-1)}{3-2}=3$, so tangent gradient $-\frac{1}{3}$. $y-2=-\frac{1}{3}(x-3)$, giving $x+3y=9$

(c) Intercepts $(9,0)$ and $(0,3)$. Area $=\frac{1}{2}\times 9\times 3=13.5$

(d) The opposite point is $(1,-4)$ (reflect P in the centre). The tangent has the same gradient: $x+3y=1-12$, so $x+3y=-11$

Question 2

(a) Substitute: $x^2+(2x+k)^2=20$, so $5x^2+4kx+k^2-20=0$. Divide by 5

(b) Tangent means $b^2-4ac=0$: $16k^2-20(k^2-20)=0$, so $k^2=100$ and $k=10$ (as $k>0$)

(c) $5x^2+40x+80=0$ gives $x=-4$, $y=2(-4)+10=2$. Point $(-4,2)$

(d) Gradient of the radius is $\dfrac{2}{-4}=-\frac{1}{2}$; the tangent has gradient 2 and $-\frac{1}{2}\times 2=-1$. A tangent is always perpendicular to the radius at the point of contact